

Sensitivity Analyses for Missing Not at Random Data in Body Donor Program Studies

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April 11, 2025

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1.1 Body Donor Program



School of Medicine and Public Health

Body Donor Program

The study of human anatomy is an essential component of medical education. Donating your body to science is a wonderful way to help provide advanced anatomy education to students entering the health professions – including medical students, physical and occupational therapy students, and physician assistant students.

- ▶ The **Body Donor Program at UW–Madison** helps train future medical professionals through anatomical education.
- ▶ Whole-body donation, rejection due to physical criteria not being met.
- ▶ My research supports this program by improving how we **predict acceptance outcomes** when donor information is incomplete.

1.2 Motivation

Goal: Improve the accuracy of body donation acceptance prediction.

Importance:

- ▶ Misclassified rejections may lead to lost donations.
- ▶ Misclassified acceptances can cause urgent funeral arrangements for families.

Key challenge: Missing data in registrants' information can bias estimates and undermine model reliability.

1.3 Missing Data Mechanisms

- ▶ **MCAR** (Missing Completely at Random):
 - ▶ Missingness has nothing to do with the data.
 - ▶ No bias introduced.
- ▶ **MAR** (Missing at Random):
 - ▶ Missingness depends on other observed variables, not the missing value itself.
 - ▶ We can adjust for it—usually doesn't bias prediction.
- ▶ **MNAR** (Missing Not at Random):
 - ▶ Missingness depends on the missing value itself.
 - ▶ Can bias results and predictions if not handled carefully.

1.5 Sensitivity Analysis

Why do we need sensitivity analysis?

- ▶ Since we can never know the true values of missing data, **any variable with missing values could, in theory, be MNAR.**
- ▶ Handle the uncertainty by exploring how our conclusions might change under different assumptions about the missingness.

What this thesis contributes:

This thesis develops a sensitivity analysis framework for **missingness in categorical predictors** (such as education and occupation), whereas most existing work focuses on missingness in outcomes.

2.1 Complete Data Prediction Model

Variables used in the model:

Variable	Description
Acceptance	Y; N
Education	$\leq$ HS; bachelors; $>$ bachelors
Occupation	17 categories
Sex	F; M
Age at Registration	Continuous
County Median Income	Continuous

Logistic regression model:

Acceptance $\sim$ Education+Occupation+Sex+County Median Income+Age at Registration

Disadvantage:

- ▶ Reduce sample size (fewer data points) $\rightarrow$ less statistical power.
- ▶ Bias: not representative of the full dataset.

2.2.1 MAR-imputed Model

MICE (Multivariate Imputation by Chained Equations):

- ▶ Imputing missing values using multiple iterations of predictive models that consider relationships between variables.

Why use MICE?

- ▶ The MAR assumption would imply that the probability of a particular variable being missing depends only on the observed values.

Missing data pattern: (1 = observed, 0 = missing)

Acceptance	Sex	Age	County Median	Education	Occupation	Frequency
1	1	1	1	1	1	2017
1	1	1	1	1	0	27
1	1	1	1	0	1	27
1	1	1	1	0	0	12

2.2.2 MICE

Sex	Age	County Median	Education	Occupation
F	69	50852	$\leq HS$	A
M	82	73726	$> HS \ \& \ \leq Bachelor's$	C
F	85	52963	—	P
F	81	60352	$\leq HS$	A
M	82	54632	—	—
M	85	54375	$> Bachelor's$	S
M	86	73726	$\leq HS$	—

2.2.2 MICE

Step 1: Set initial values:

Sex	Age	County Median	Education	Occupation
F	69	50852	$\leq HS$	A
M	82	73726	$> HS \ \& \ \leq Bachelor's$	C
F	85	52963	$\leq HS$	P
F	81	60352	$\leq HS$	A
M	82	54632	$\leq HS$	A
M	85	54375	$> Bachelor's$	S
M	86	73726	$\leq HS$	A

2.2.2 MICE

Step 2: Set variable of interest back to missing (Occupation):

Sex	Age	County Median	Education	Occupation
F	69	50852	$\leq HS$	A
M	82	73726	$> HS \ \& \ \leq Bachelor's$	C
F	85	52963	$\leq HS$	P
F	81	60352	$\leq HS$	A
M	82	54632	$\leq HS$	—
M	85	54375	$> Bachelor's$	S
M	86	73726	$\leq HS$	—

2.2.2 MICE

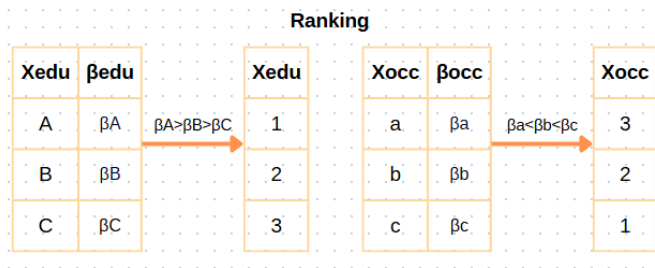
Step 3: Build a regression model of Occupation predicted by the other 4 variables, using cases where Occupation was observed:

Sex	Age	County Median	Education	Occupation
F	69	50852	$\leq HS$	A
M	82	73726	$> HS \ \& \ \leq Bachelor's$	C
F	85	52963	$\leq HS$	P
F	81	60352	$\leq HS$	A
M	82	54632	$\leq HS$	—
M	85	54375	$> Bachelor's$	S
M	86	73726	$\leq HS$	—

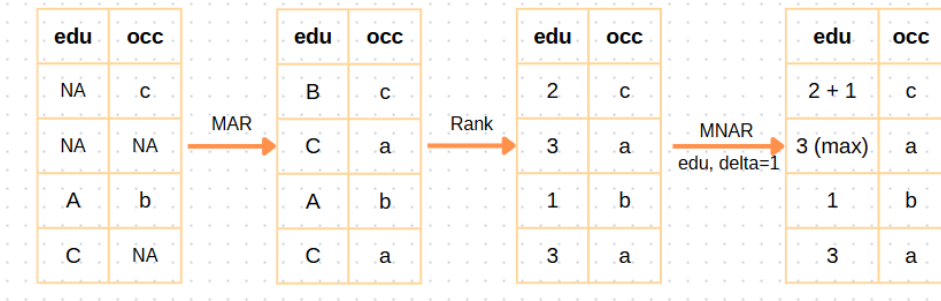
Step 4: Predict the missing Occupation values using the model.

2.3 Ranking

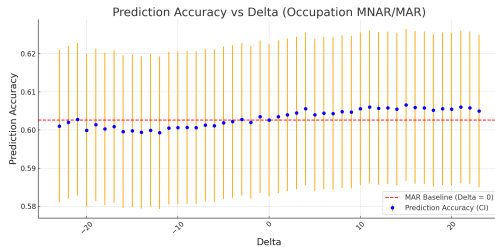
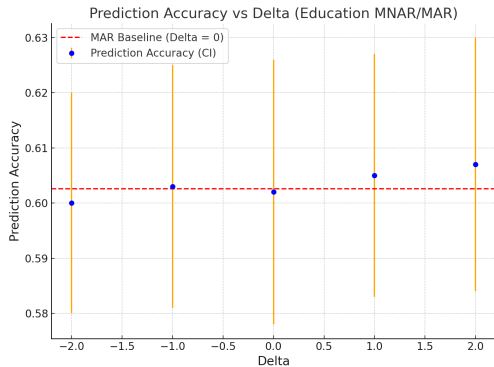
- ▶ Ranks are assigned to the categorical variables Education and Occupation based on the order of their regression coefficients in the MAR-imputed model.
- ▶ Purpose: Ranking allows us to systematically simulate MNAR scenarios by deviating from MAR-imputed values.



2.4 MNAR Sensitivity Analysis



3 Results



Takeaway: Sensitivity Analysis enables to display all possible outcomes when we are not sure about missing data mechanisms.

Acknowledgement & References

I would like to thank my supervisor, Professor Rick Chappell, and many others for their feedback and support on this project. I acknowledge the UW-Madison L&S Honors Program for funding my research through the Trewartha Senior Thesis Research Grant.

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